Geospatial Data Analysis

Sales Visualization, Geographic Clusters and Underserved Market Detection

Project Type: Cloud/Data Analytics practical project

Visualization Tool: Python + Folium

Dataset: Synthetic demonstration dataset created for this submission because no source dataset was supplied.

1. Introduction

Geospatial data analysis studies information together with its geographic location. In this project, regional sales and user information are analyzed using State, City, Postal Code and geographic coordinates. The purpose is to understand where demand is concentrated and where a business may have an opportunity to expand.

2. Objective

The main objective is to visualize sales data geographically, identify high-demand geographic clusters, and screen for potentially underserved regions. The analysis combines revenue, user count and revenue-per-user into a simple opportunity score.

3. Dataset Description

The demonstration dataset contains 20 city-level records across 13 states. Important fields are State, City, Postal Code, Revenue, Users, Latitude and Longitude. Revenue is treated as sales value in Indian Rupees.

4. Data Cleaning

Duplicate records were removed and rows missing essential location, revenue, user or coordinate values were excluded. Revenue and Users were converted to numeric data types. Derived fields were then calculated for revenue per user and demand index.

5. Regional Aggregation

Revenue and user counts were aggregated by State. Revenue per user was calculated as total revenue divided by total users. This provides a simple comparison of commercial performance and user concentration between regions.

6. Spatial Analysis Method

An interactive Folium map was created using city latitude and longitude. Circle/marker information shows local metrics, while a heat layer represents concentrations of users. A screening score combines high user demand with relatively low revenue per user. A higher score indicates a region worth investigating further as a potential underserved market.

7. High-Demand Geographic Clusters

The heat layer makes it easier to visually identify areas with larger concentrations of users. Large metropolitan areas in the demonstration data naturally appear as stronger demand centers. This visualization should be interpreted together with revenue and competitive coverage rather than on its own.

8. Top 3 High-Potential Underserved Regions

Based on the demonstration scoring method, the following three city-level locations receive the highest opportunity scores. These are analytical candidates, not confirmed underserved markets.

Rank	State	City	Users	Revenue	Revenue/User	Score
1	Bihar	Patna	2,500	■210,000	■84.00	0.600
2	Uttar Pradesh	Lucknow	2,800	■390,000	■139.29	0.575
3	Odisha	Bhubaneswar	2,200	■180,000	■81.82	0.550

9. Recommendations

The business should validate the shortlisted regions using competitor presence, population, purchasing power, service availability, logistics cost, customer acquisition cost and local demand trends. Pilot expansion can be tested in one or two locations before a larger investment.

10. Limitations

The supplied project did not include a real dataset, so this report uses synthetic demonstration data. Geographic coordinates are approximate city-center coordinates. The underserved score is a screening metric and does not measure actual market share or service availability. Real decisions require real operational and market data.

11. Conclusion

Geospatial analysis provides a practical way to combine sales and customer information with location. The interactive map helps decision-makers see demand patterns, while the scoring method creates a transparent shortlist of potential expansion areas. With real regional data, the same workflow can be reused to produce evidence-based expansion recommendations.

12. References

Folium documentation; Python pandas documentation; geographic visualization concepts; cloud/data analytics course materials.

Appendix: Submission Files

The accompanying HTML file contains the interactive map, and the Jupyter Notebook contains the reproducible analysis workflow. Replace the demonstration dataset with the actual project dataset before using the results for real business decisions.